Haifeng Sun

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Education

Shanghai Jiao Tong University September 2024 – present
B.Eng in robotics and automation Shanghai, China

- GPA 3.7/4.0, Average 87/100
- **research interest**: Robot Learning, Manipulation, SLAM, and Human-Robot Interaction.

Research Experience

Machine Vision and Intelligence Group May 2025 – Now
Supervisor: Prof. Yong-Lu Li and Prof. Cewu Lu Shanghai, China

- Research dirction: Imitation learning, multimodal data, transfer learning, and snake-like robot control.
- Leader of robopanoptes — The All-Seeing Robot with Whole-body Dexterity.
- Proposing a novel idea of **progressive training** for multi-degree-of-freedom control.

IWIN-FINS Lab November 2024 – May 2025
Supervisor: Prof. Jianping He Shanghai, China

- Research dirction: Embedded systems, control theory, and SLAM.
- One of the top contributors of FinesRobot — a fully self-developed, advanced steerable wheeled mobile robot equipped with the Finemote embedded framework system.
- Designer of **Finemote** - a modular, real-time embedded framework that simplifies the development of high-mobility robotics applications.

Projects

robopanoptes 2025 – Now
Leader RHOS/robopanoptes

- Led the overall development of RoboPanoptes, a robotic system integrating whole-body vision and diffusion-based visuomotor policies.
- Built and maintained the robot hardware platform.
- Designed and implemented a whole-body vision-based learning framework, leveraging CLIP for multi-modal perception anddiffusion models for visuomotor policy learning.

Finemote-Fines-Robot 2024 – Now
Embedded System and Control Designer IWIN/Fines

- Designed and implemented the Finemote embedded framework, a layered and modular real-time control system providing hardware abstraction.
- Developed both the hardware and software algorithms for the high-precision swerve-drive chassis.
- Built an anti-slip control strategy by combining **MPC** and **Sliding Mode Control** to ensure stability on challenging terrains.

Skills

Programming Languages: C, C++, Python, Matlab and JavaScript
Tech Skills: STM32, ROS, Arduino, Pytorch, Solidworks

References

Cewu Lu: Professor of AI/CS, Shanghai Jiao Tong University
Yonglu Li: Assistant Professor of AI/CS, Shanghai Jiao Tong University
Jianping He: Professor of robotics/automation, Shanghai Jiao Tong University